WISER 2026
INDUSTRY CHALLENGE

QUANTUM OPTIMIZATION FOR DISTRIBUTED ORDER MANAGEMENT

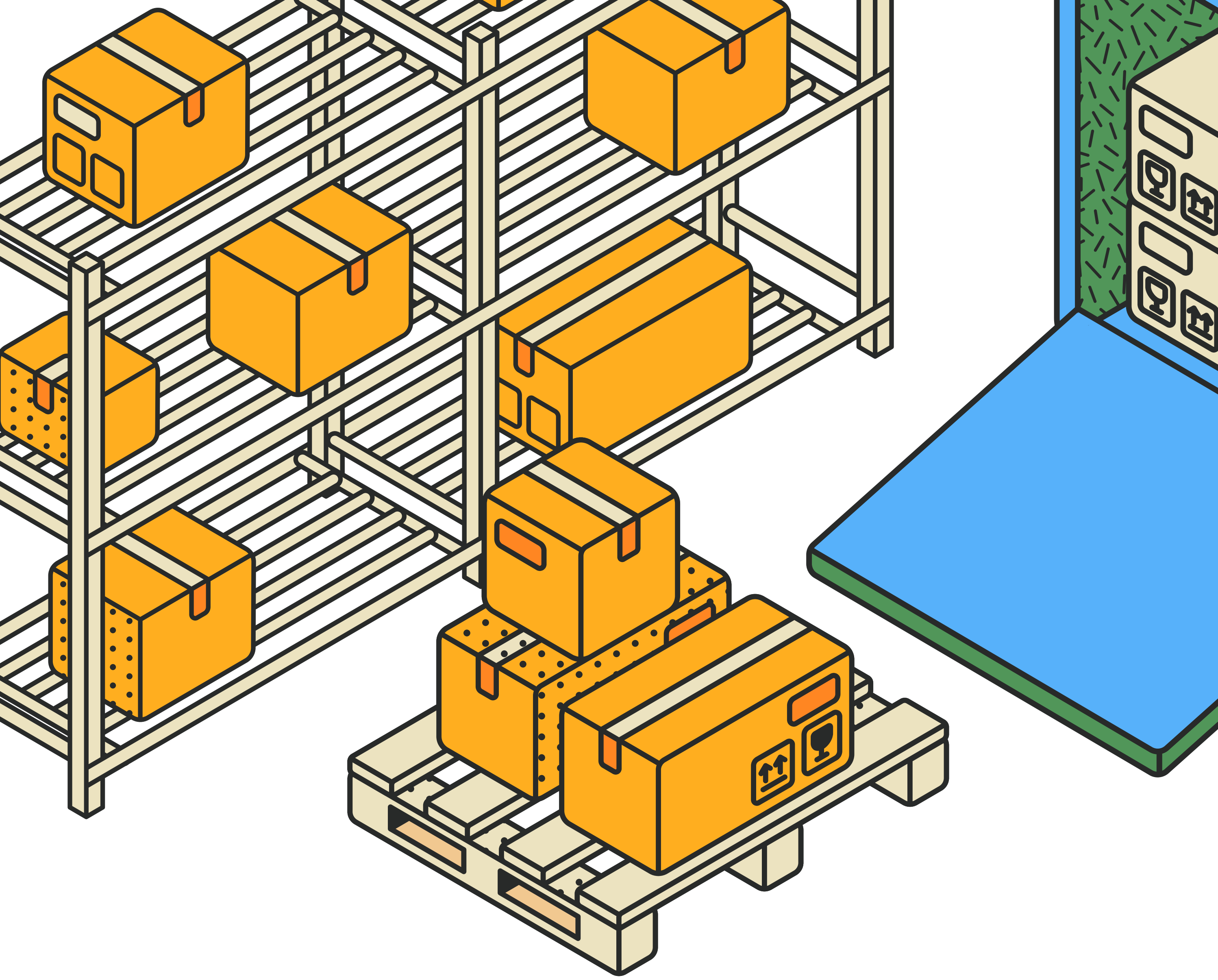
NESTLÉ CHALLENGE

Entangled Minds

ENTANGLED MINDS

THE BUSINESS PROBLEM

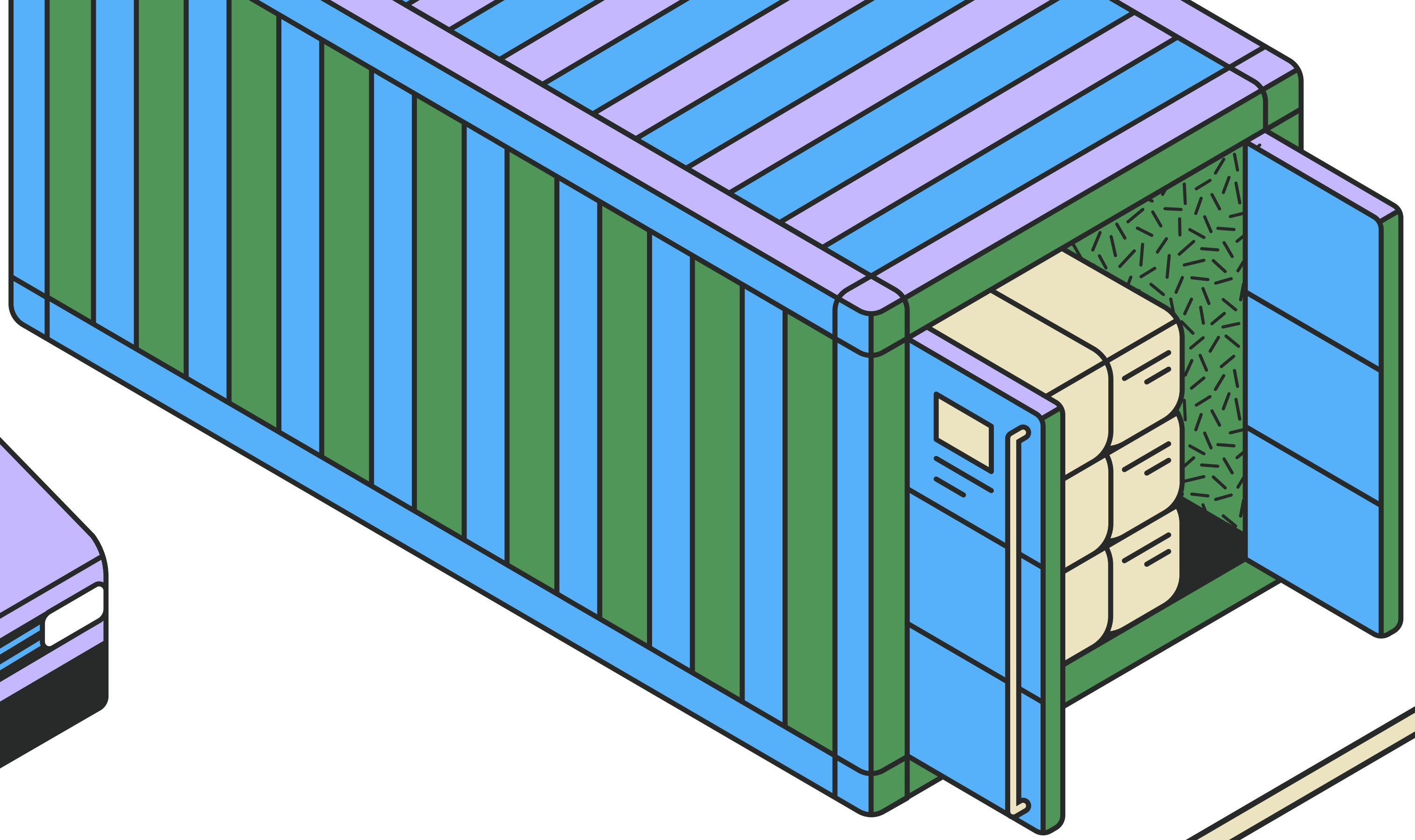
Nestlé's distribution network handles thousands of orders daily. When the assigned warehouse runs out of stock planners have to decide where to redirect the order. The goal of Distributed Order Management (DOM) is to automate and optimize the reassignment decisions, balancing recovery value against shipping costs and penalties.



OUR APPROACH

Our evaluation framework is built on three layers to ensure fairness.

- Classical Baselines:
  - Default Assignment: No optimization.
  - Greedy Heuristic: Simple sequential reassignment.
- Exact Optimization:
  - Integer Linear Programming (ILP) solved with CBC, guaranteeing optimality for the tractable subset.
- Quantum & Quantum-Inspired:
  - Quantum-Inspired: Simulated Annealing, Path-Integral QMC, Tabu Search, Steepest Descent (on QUBO).
  - Real Quantum: A gate-based QAOA circuit (Qiskit-Aer) on a reduced 6-qubit instance.



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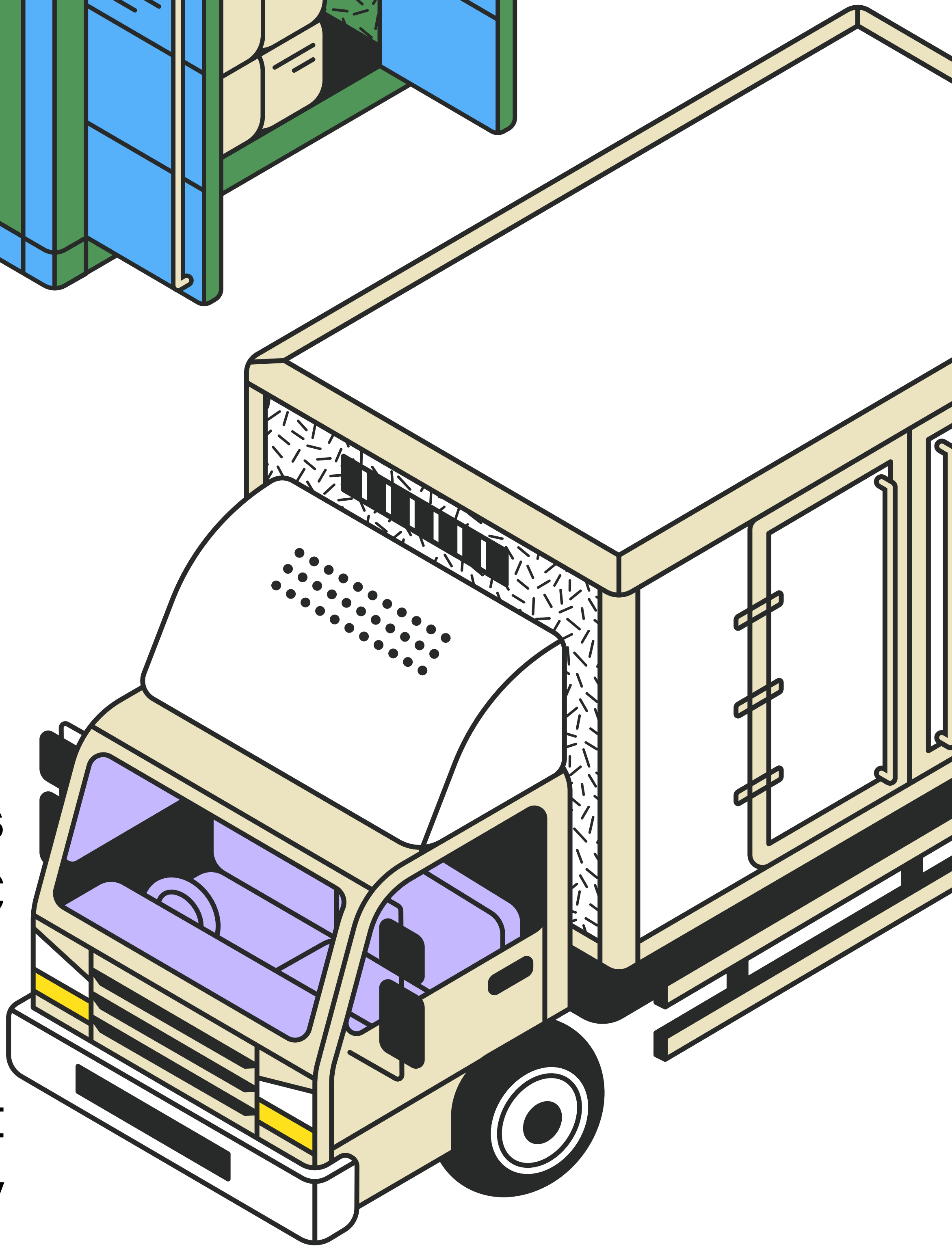
THE OPTIMIZATION FORMULATION



We use a binary variable x to decide if an order goes to a specific DC.

Our objective is a single equation that captures everything a planner cares about: capturing order revenue and subtracting shipping costs for the DC we choose, while subtracting penalties if we can't fulfill the order.

We then impose three hard constraints: we can't split an order, we can't exceed daily inventory per SKU, and we can't exceed the warehouse's daily throughput capacity.

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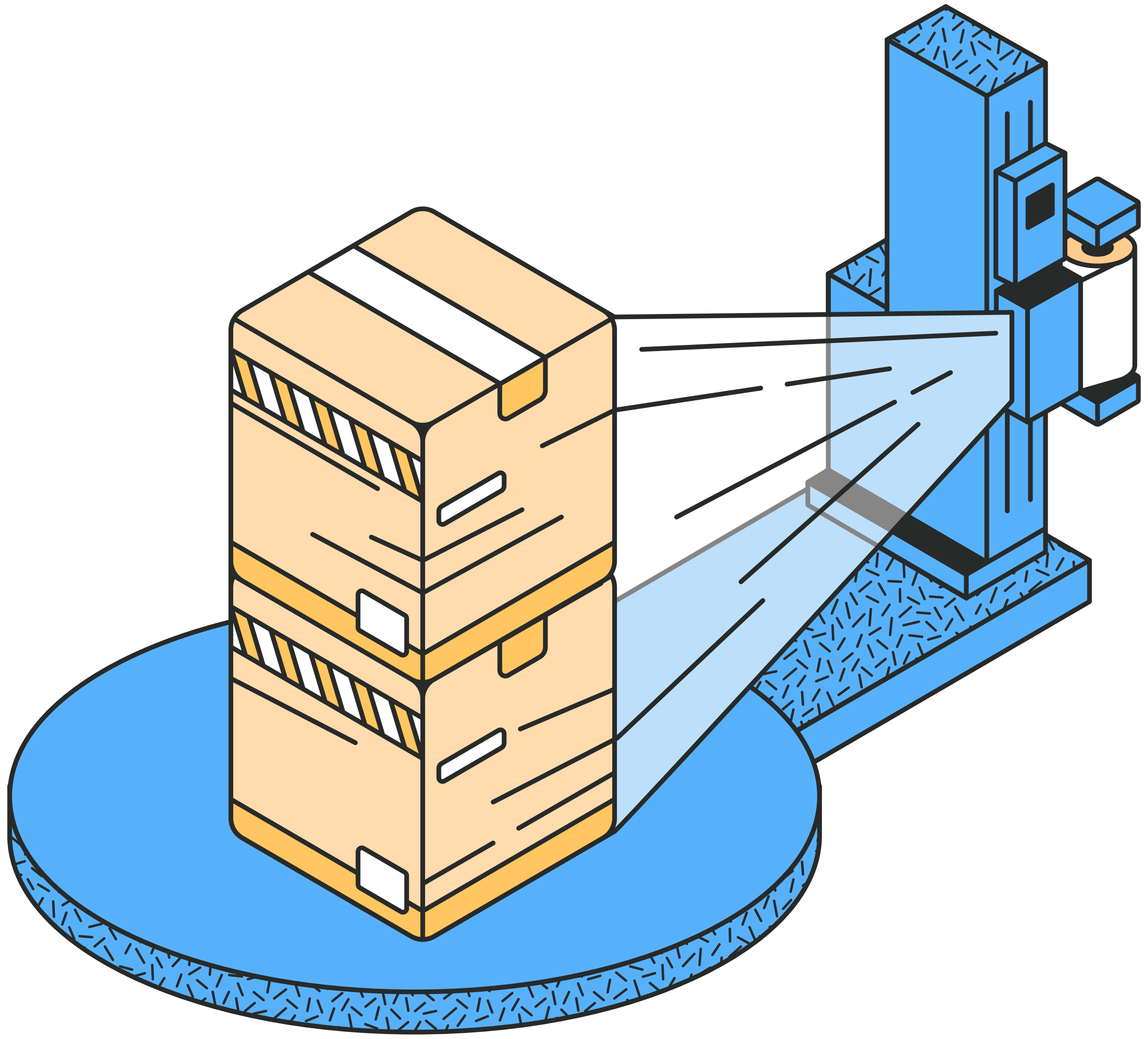
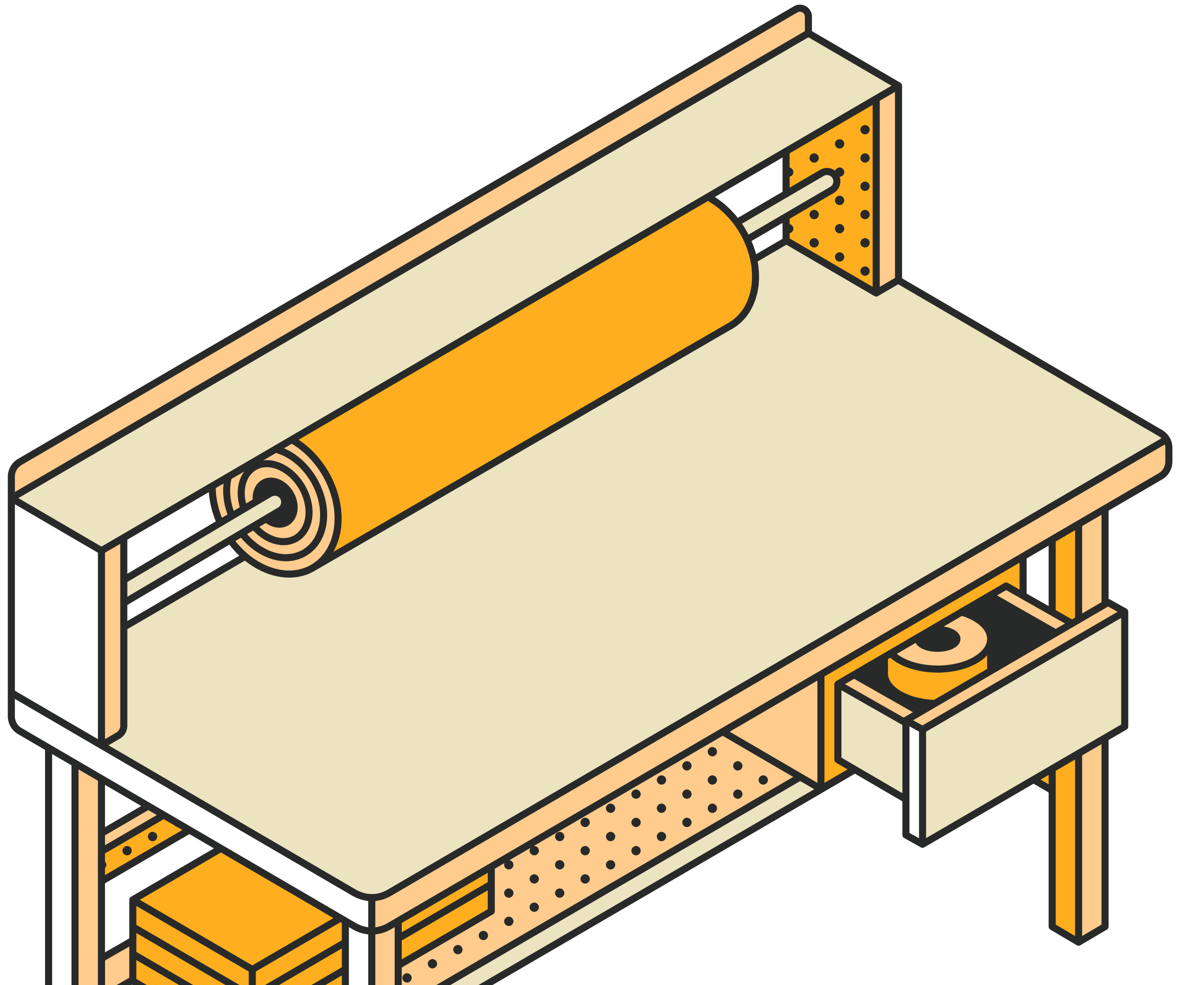
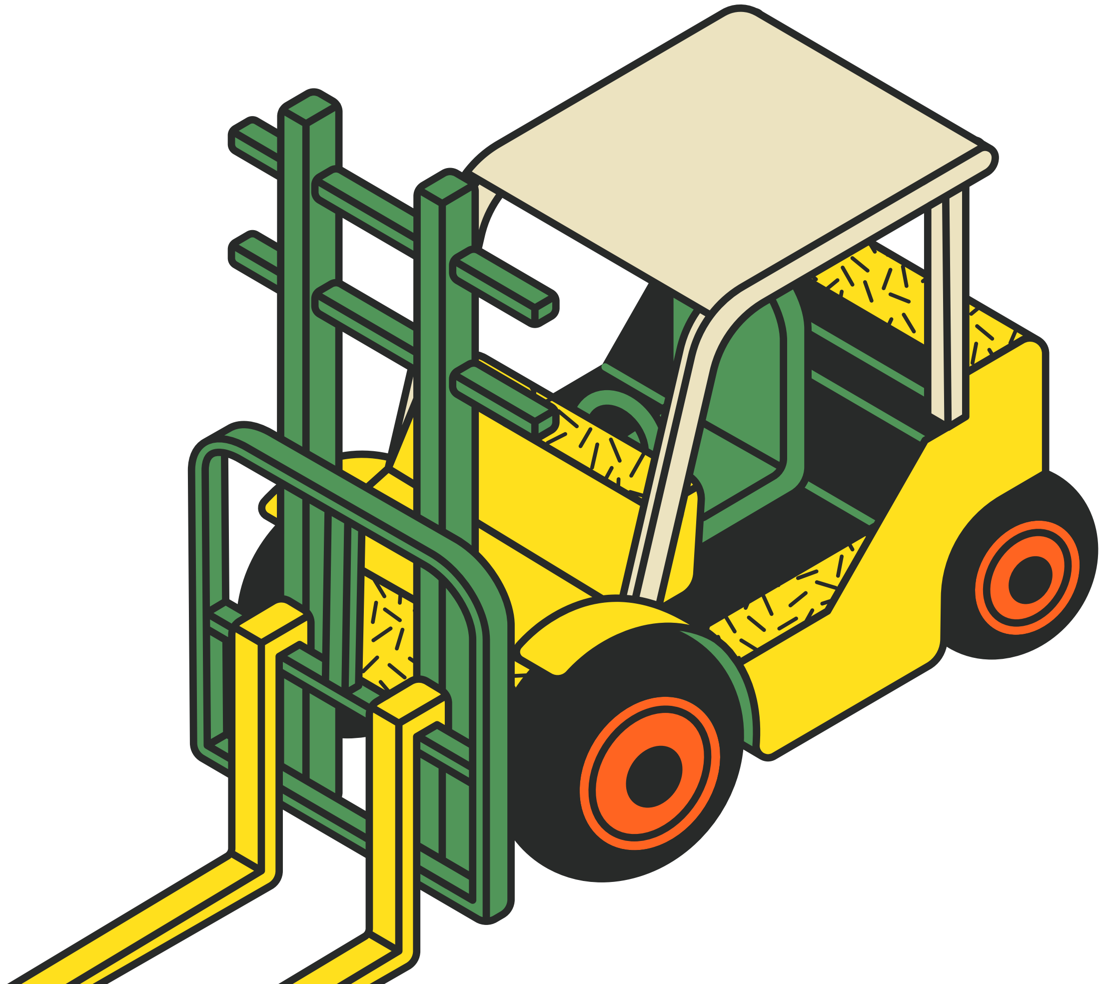
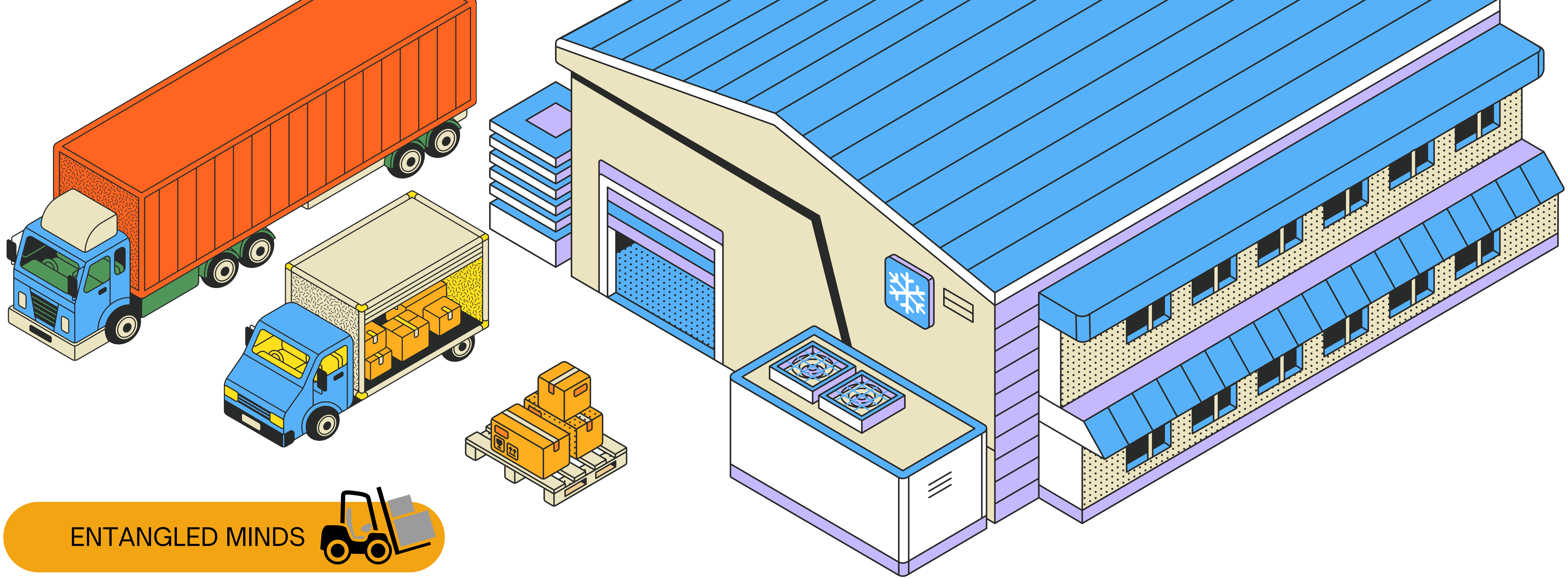
KEY RESULTS

Method	Objective Value	Fill Rate	Reassigned Orders	Shipping Cost	Penalty Cost
Default Assignment	229,995	14%	0	12,175	15,725
Greedy Heuristic	412,060	89%	72	171,027	3,740
ILP (Optimal)	447,822	53%	38	89,434	6,036









QUANTUM-INSPIRED METHODS

Method	Objective	% of ILP Optimum	Fill Rate	Runtime (s)
Random Sampling	-8,616	—	2%	0.003
Simulated Annealing (SA)	369,752	82.60%	71%	3.2
Path-Integral QMC	358,764	80.10%	69%	2.2
Tabu Search	439,188	98.10%	53%	100.2
Steepest Descent (QUBO)	444,071	99.20%	53%	0.08

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SCALING, LIMITATIONS & THE ROAD AHEAD

Scope	Orders	DCs	Variables
QAOA Subset (Solved)	3	2	6
Tractable Subset (Solved)	93	8	744
All Focus Orders	359	8	2,872
All Open Orders	614	8	4,912
Full Multi-SKU Model	25,193 SKU lines	8	~2,000,000+